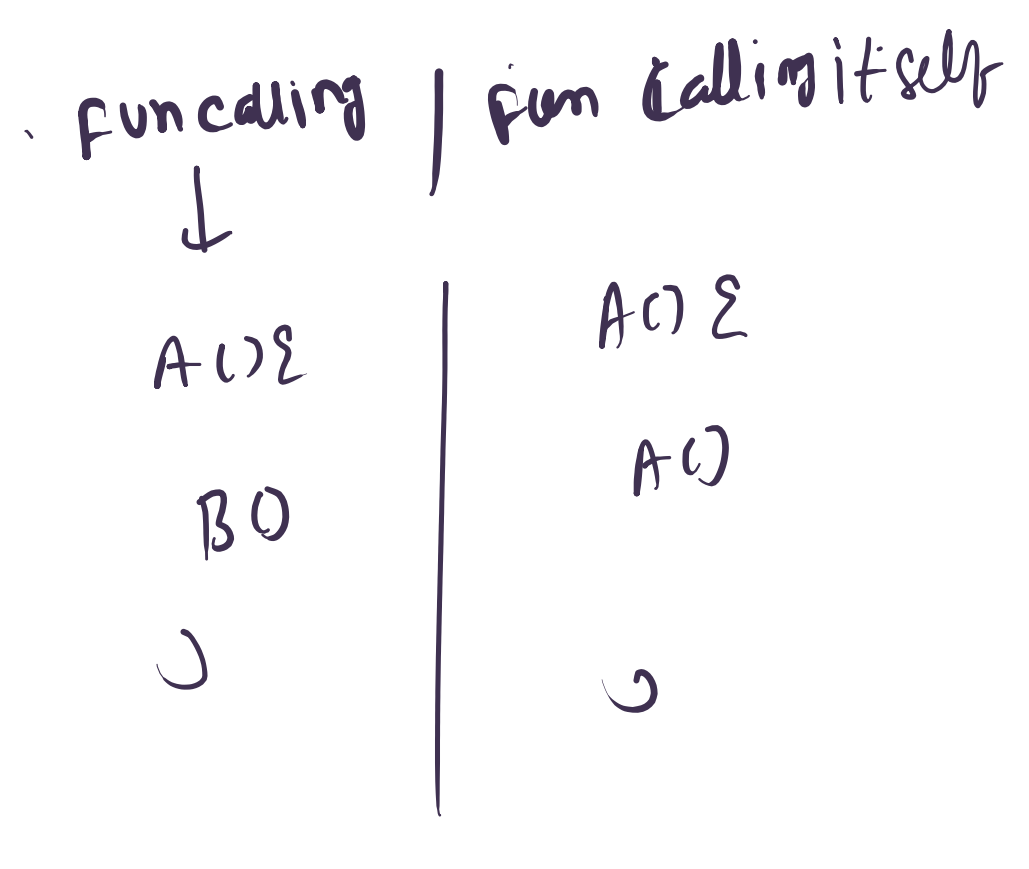
Lec-18
15/06/26 7:14 PM

Recursion



PMI Math

① $\sum_{i=1}^{N} i = 1+2+3+\dots N = \frac{N\times(N+1)}{2}$

$\sum_{i=1}^{1} i = 1 = \frac{1\times(1+1)}{2}$

$\sum_{i=1}^{2} i = 1+2 = \frac{2\times(2+1)}{2}$

② $\sum_{i=1}^{K} i = 1+2+3+\dots+K = \boxed{\frac{K(K+1)}{2}}$

① Smaller value
② N=K correct
③ N=K+1 →

③ $\sum_{i=1}^{K+1} i = 1+2+3+4 \dots K+K+1 = \frac{(K+1)(K+2)}{2}$

$= \frac{K(K+1)}{2} + (K+1) =$

$= (K+1)\left[\frac{K}{2}+1\right] =$

$= \frac{(K+1)(K+2)}{2}$

Recursion 4 दिन

last Sacha

Function Calling itself

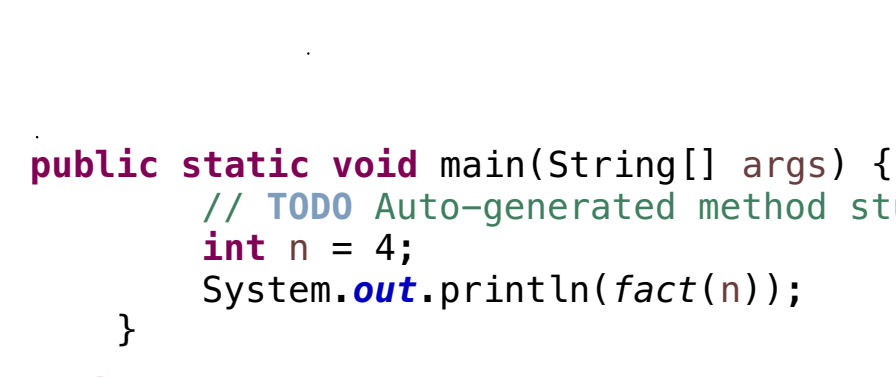
3. Step
① n-1 → Recursion
② Result use → original Problem
③ Base-case → PMI-1st

# $5! = \underbrace{4!}_{24} \times 5$ (120)

6 ← 3! × 4
2 ← 2! × 3
1 ← 1! × 2
1 ← 0! × 1

Code
way code

(N-1)

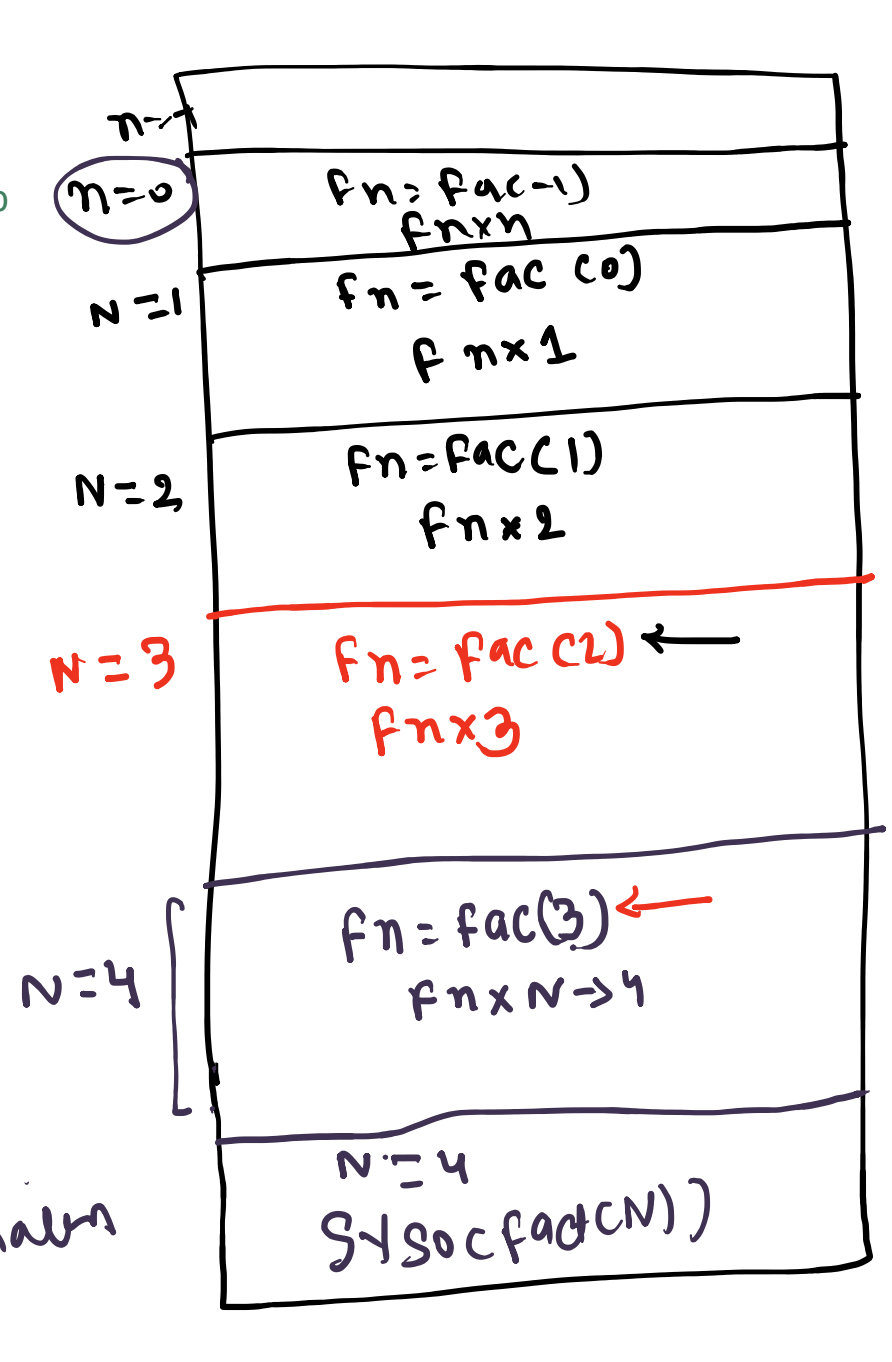


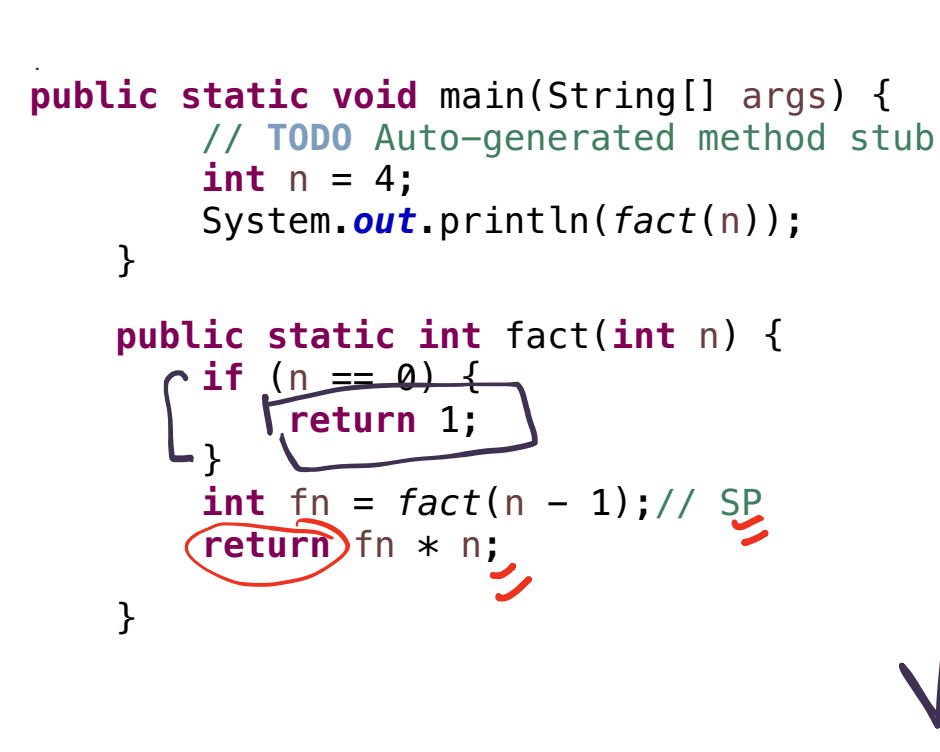
```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 4;
        System.out.println(fact(n));
    }

    public static int fact(int n) {

        int fn = fact(n - 1);// SP
        return fn * n;

    }
```





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public static void main(String[] args) {
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        int n = 4;
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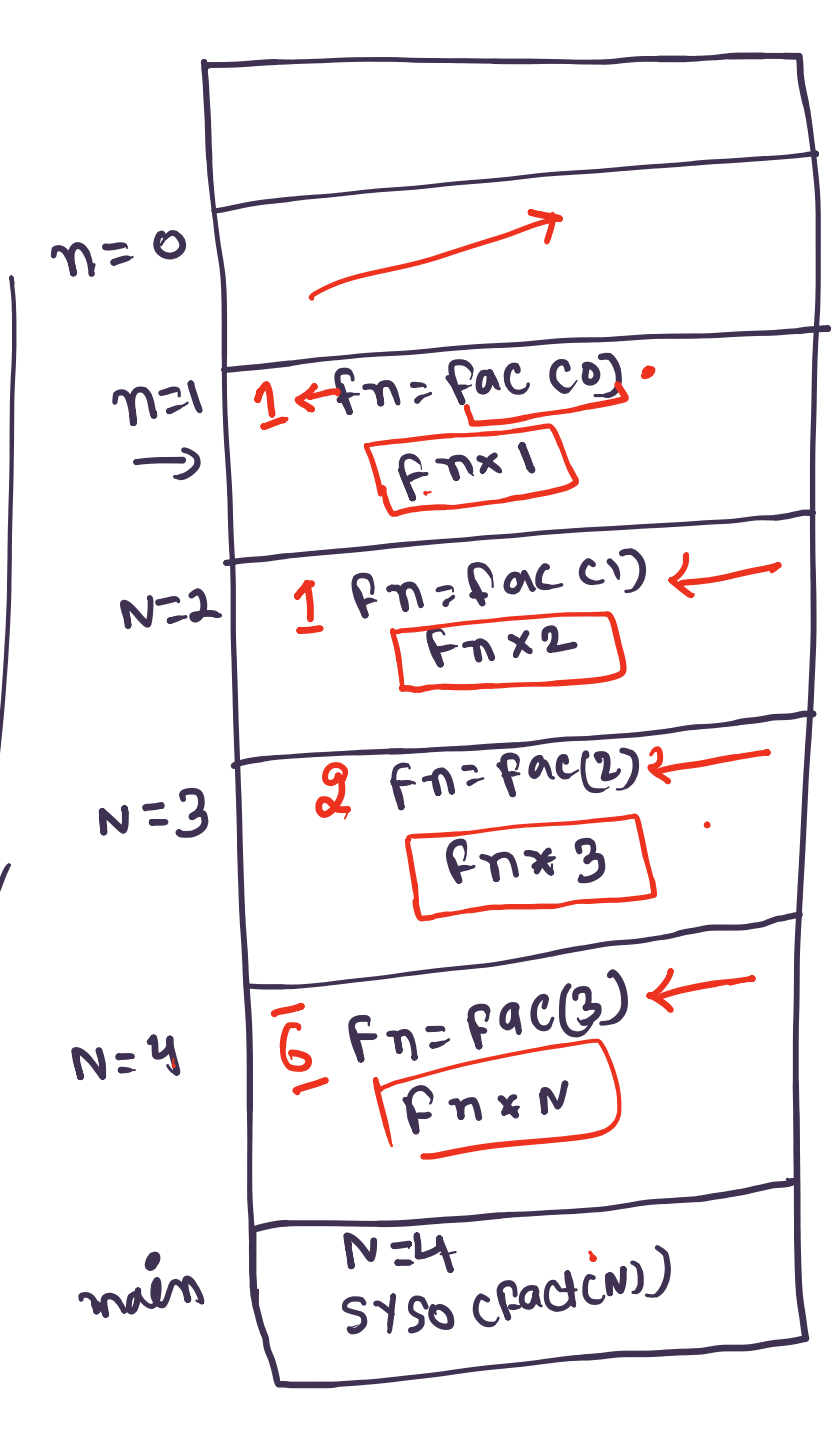
    }

    public static int fact(int n) {
        if (n == 0) {
            return 1;
        }
        int fn = fact(n - 1);// SP
        return fn * n;
    }
```

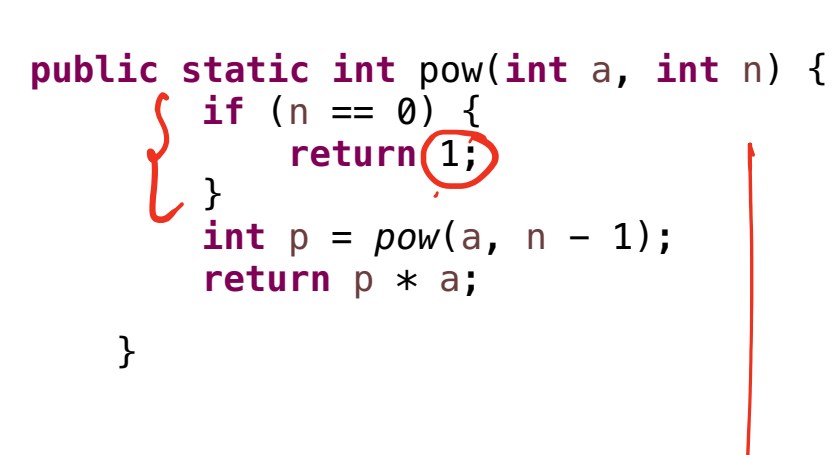


$a^n \rightarrow [a^{n-1}] \times a$

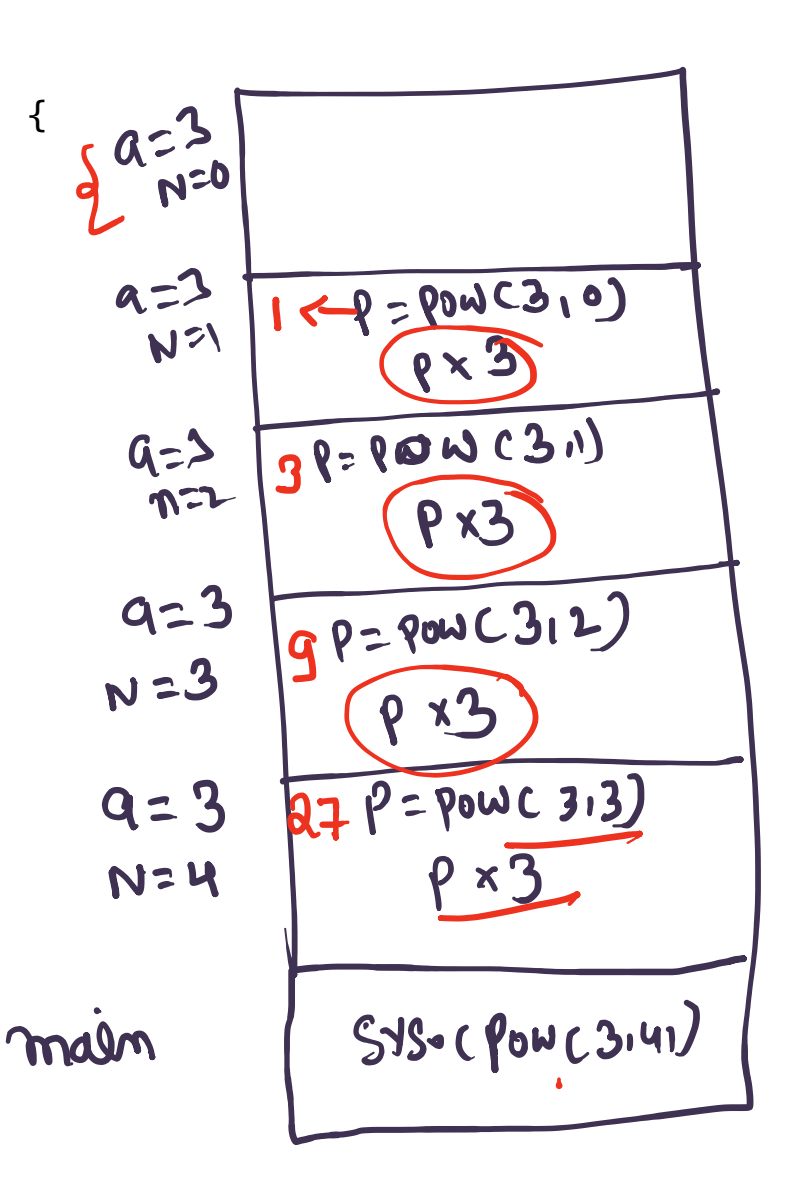
n==0
①

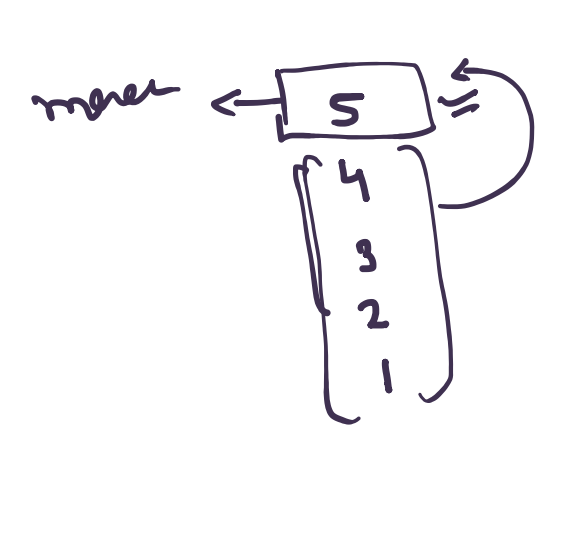
int P = pow(a, n-1)

(p×a)



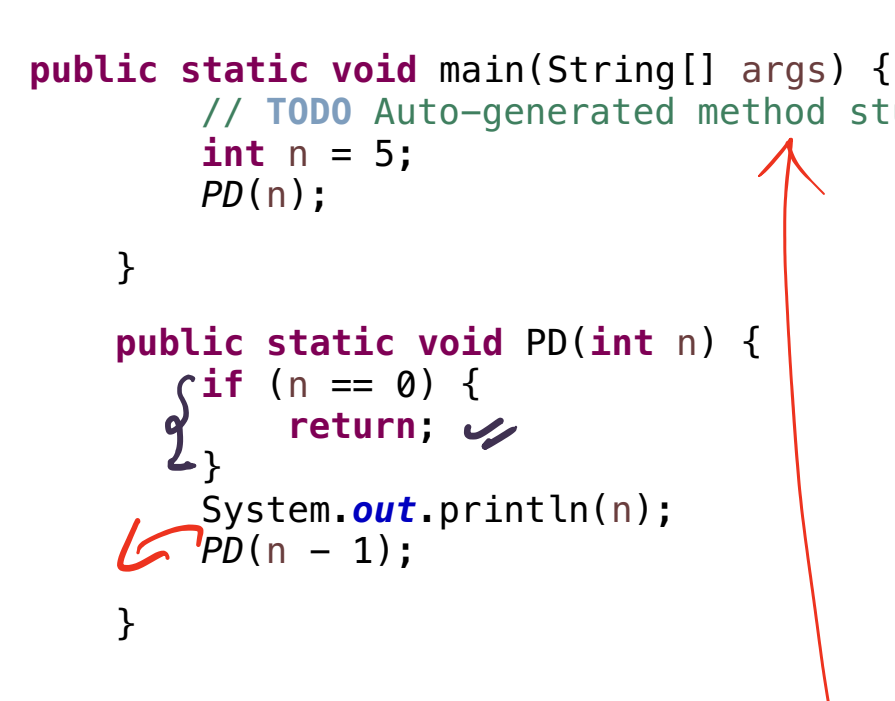
```java
public static int pow(int a, int n) {
        if (n == 0) {
            return 1;
        }
        int p = pow(a, n - 1);
        return p * a;

    }
```



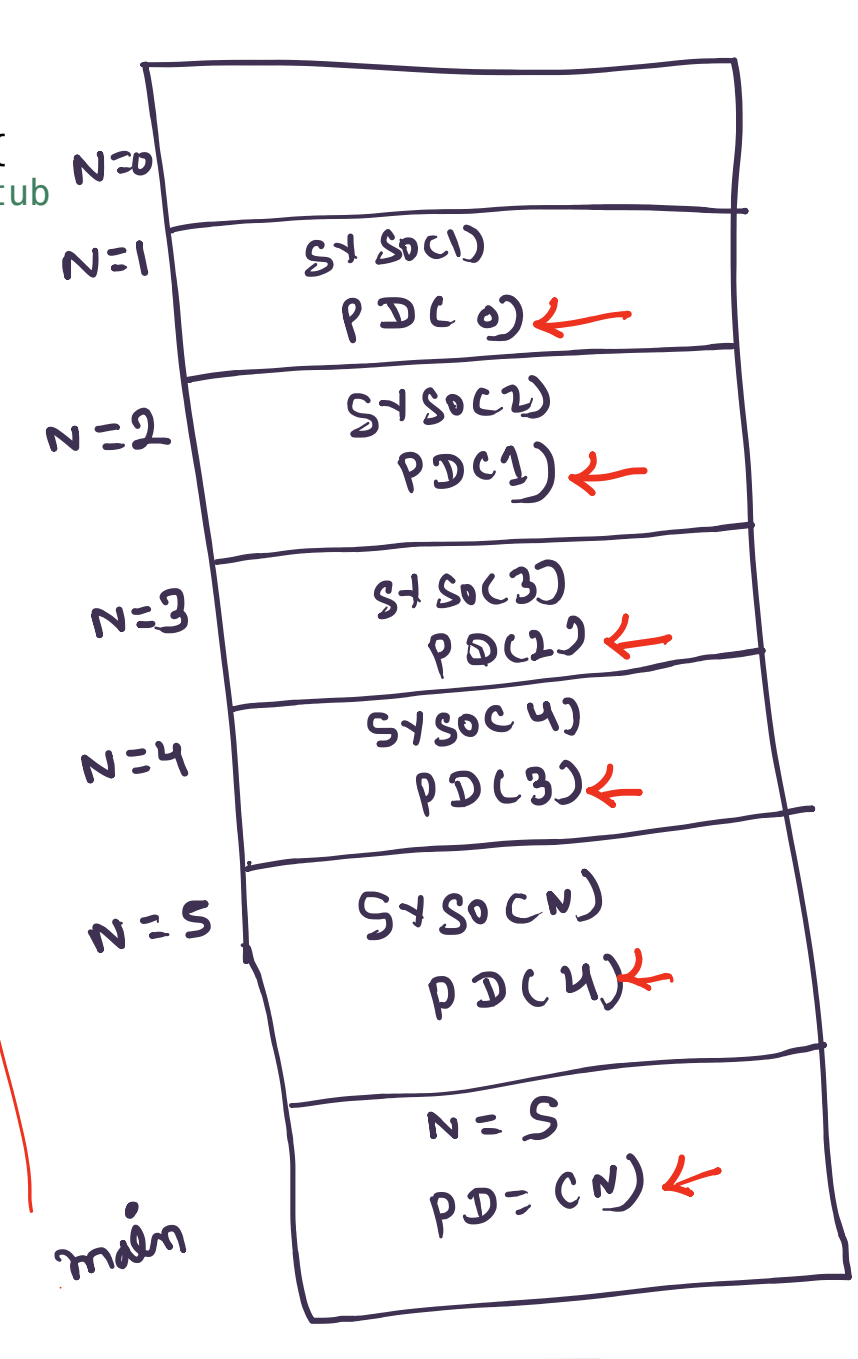


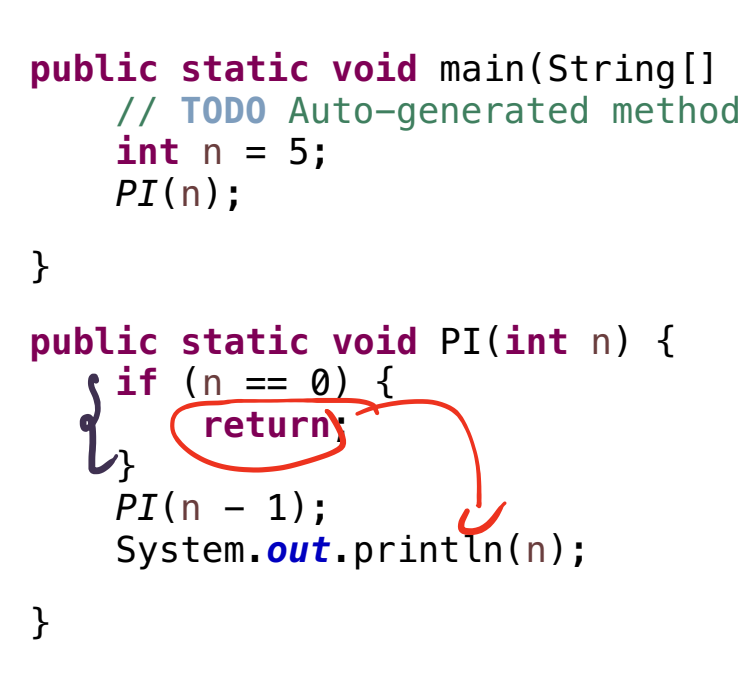
N=5

PD(N-1)

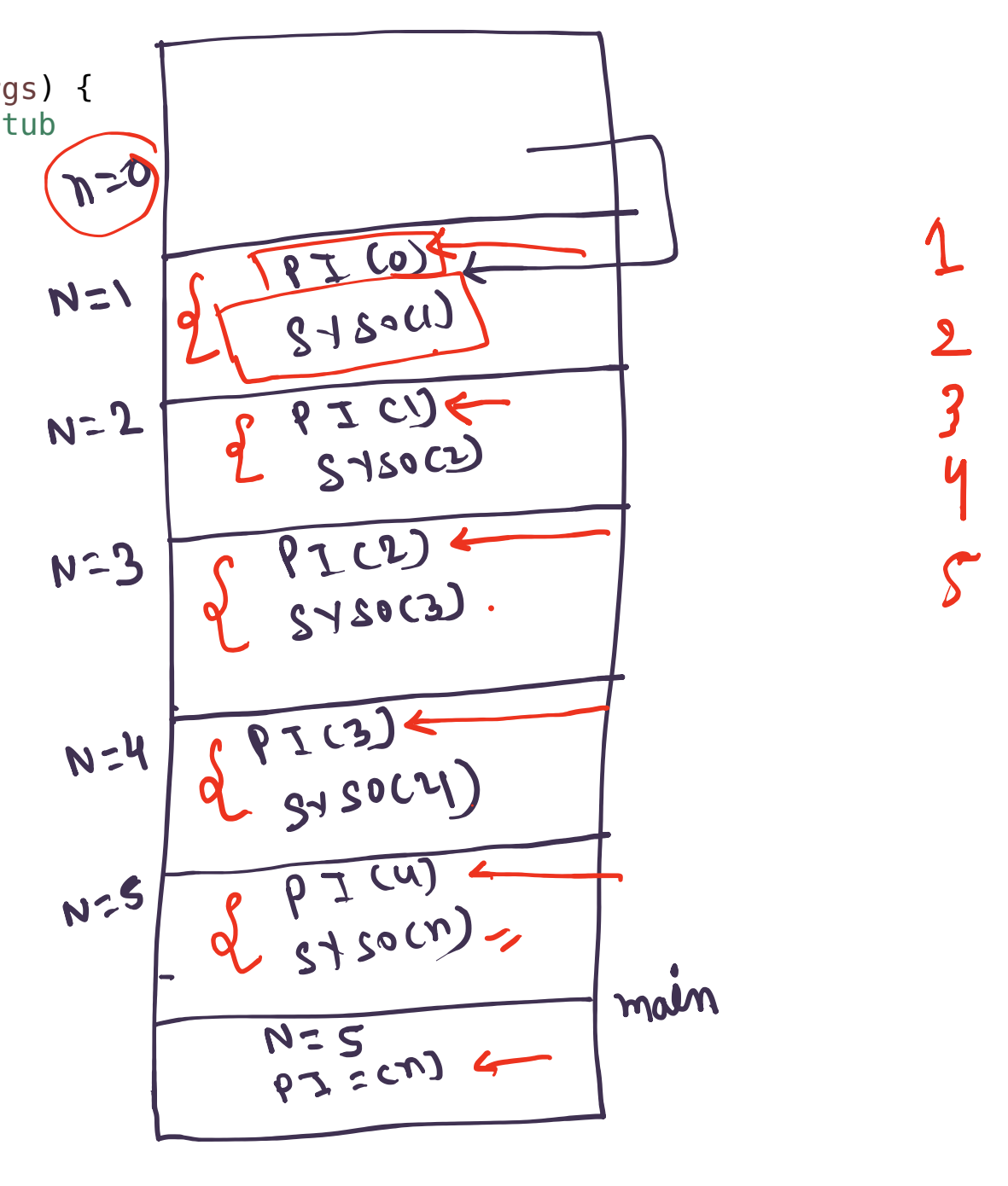


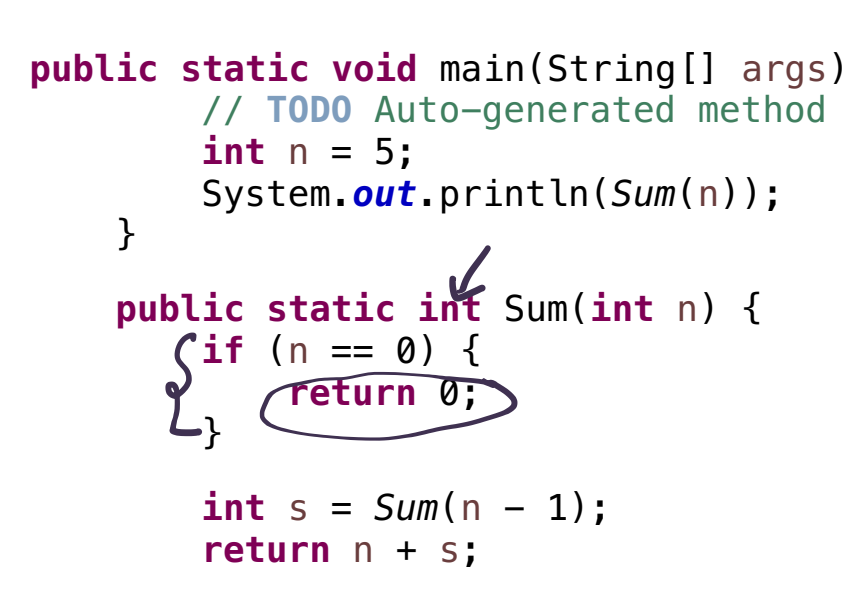
```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        PD(n);

    }

    public static void PD(int n) {
        if (n == 0) {
            return;
        }
        System.out.println(n);
        PD(n - 1);

    }
```



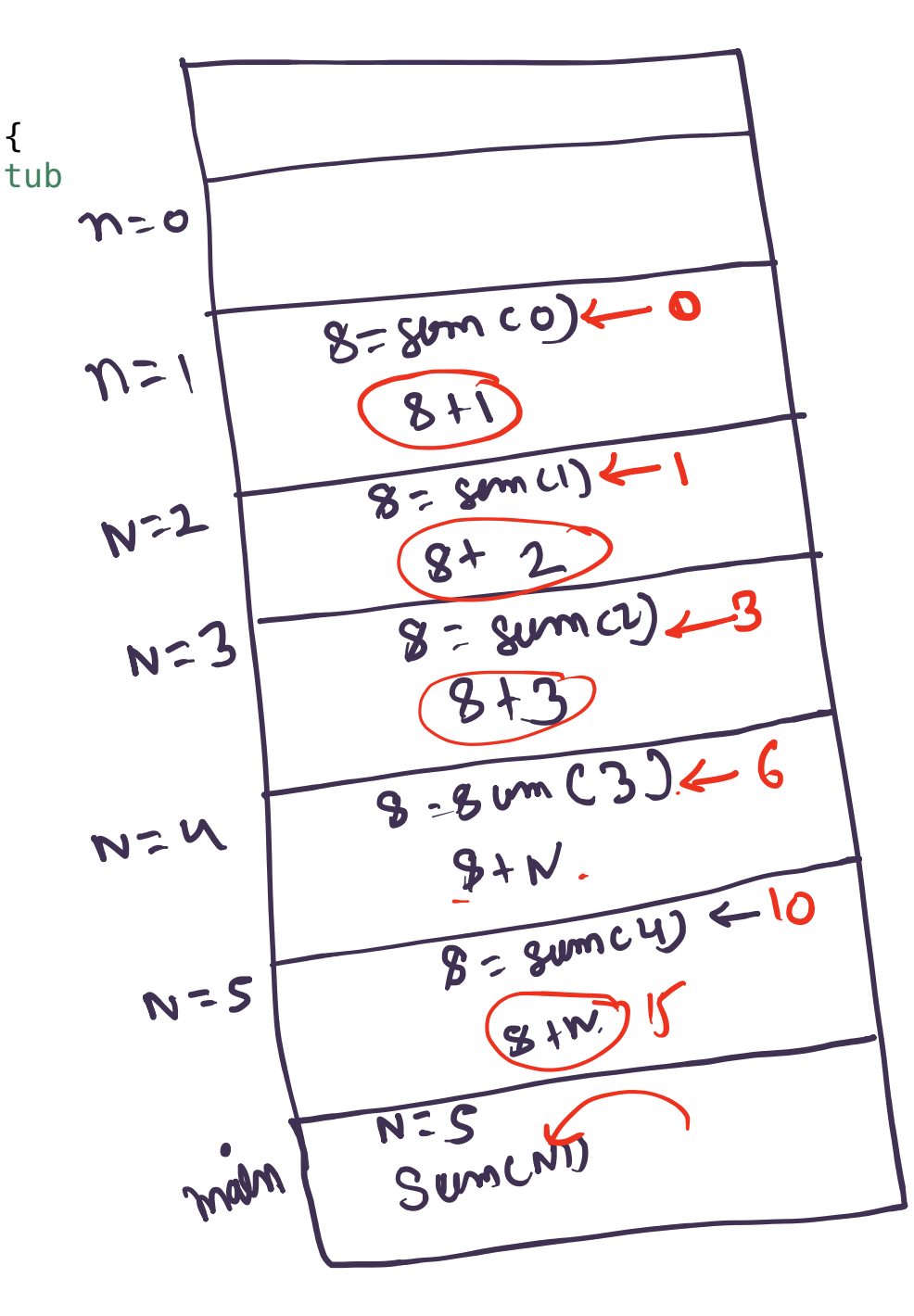


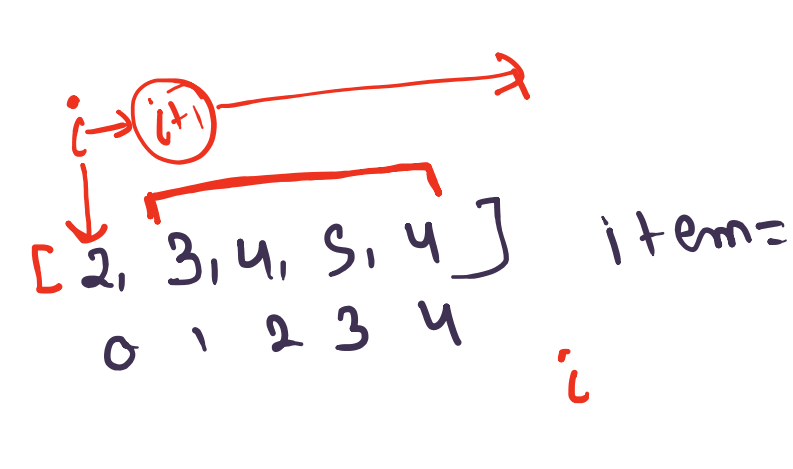
```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        PI(n);

    }

    public static void PI(int n) {
        if (n == 0) {
            return;
        }
        PI(n - 1);
        System.out.println(n);

    }
```

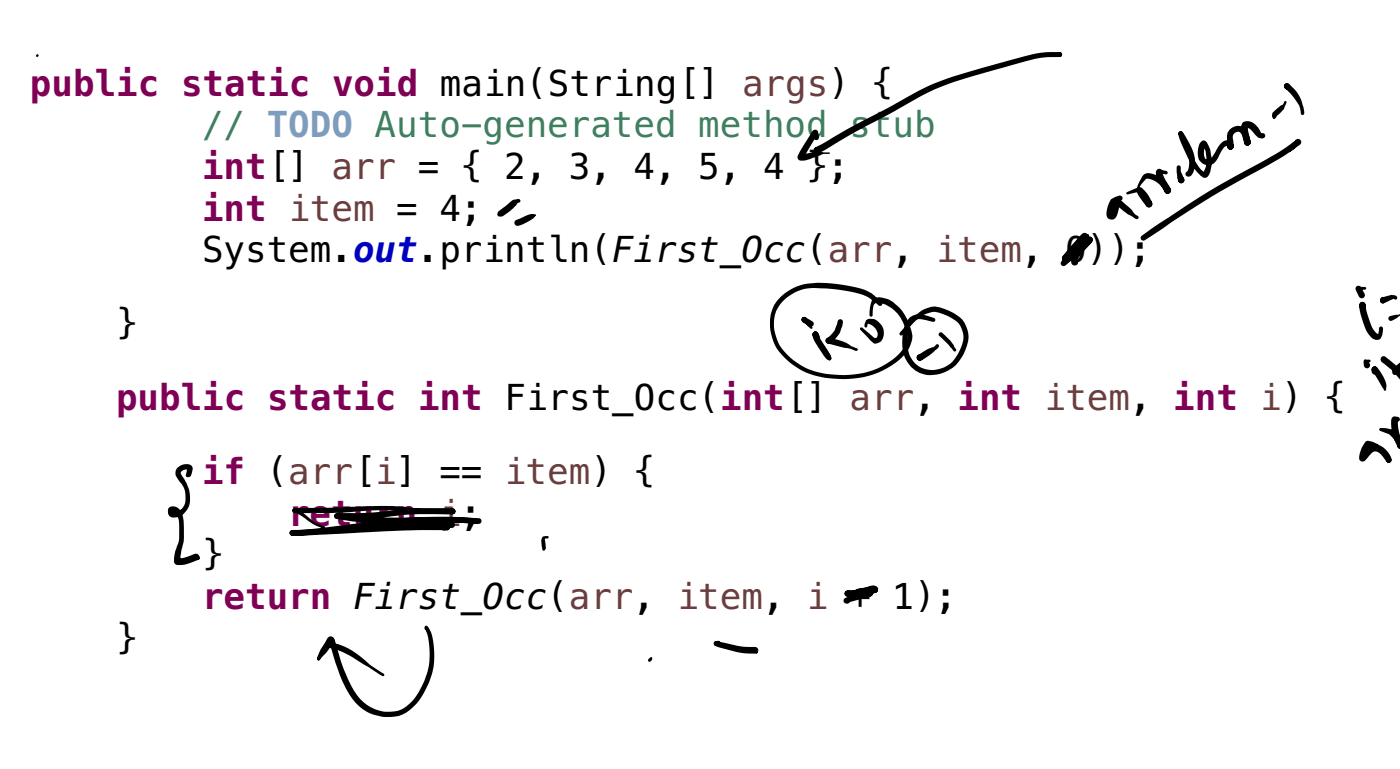




```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        System.out.println(Sum(n));
    }

    public static int Sum(int n) {
        if (n == 0) {
            return 0;
        }

        int s = Sum(n - 1);
        return n + s;

    }
```



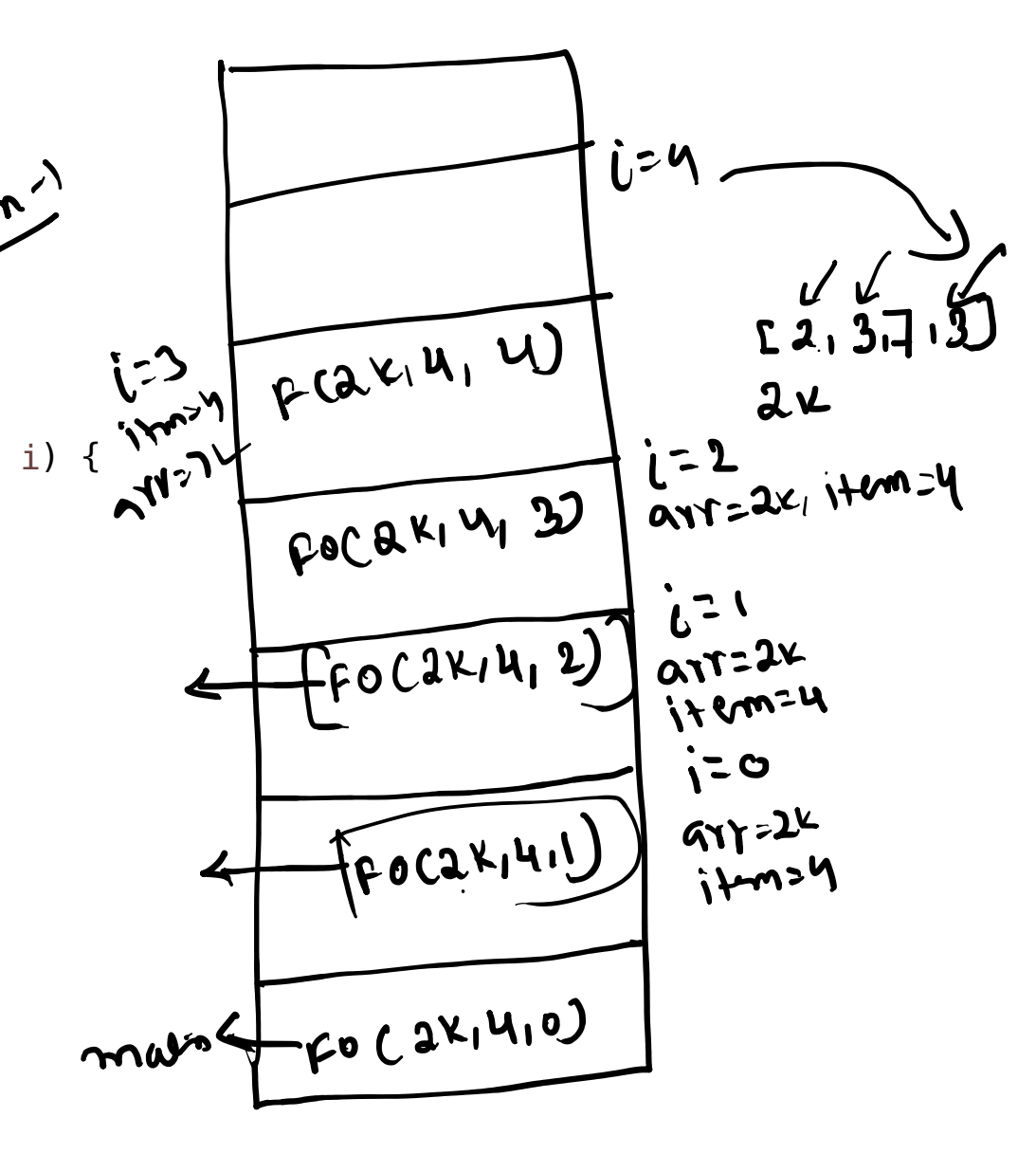




```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int[] arr = { 2, 3, 4, 5, 4 };
        int item = 4;
        System.out.println(First_Occ(arr, item, 0));

    }

    public static int First_Occ(int[] arr, int item, int i) {

        if (arr[i] == item) {
            return [illegible];
        }
        return First_Occ(arr, item, i + 1);
    }
```

arr.len-1

i<0 -1



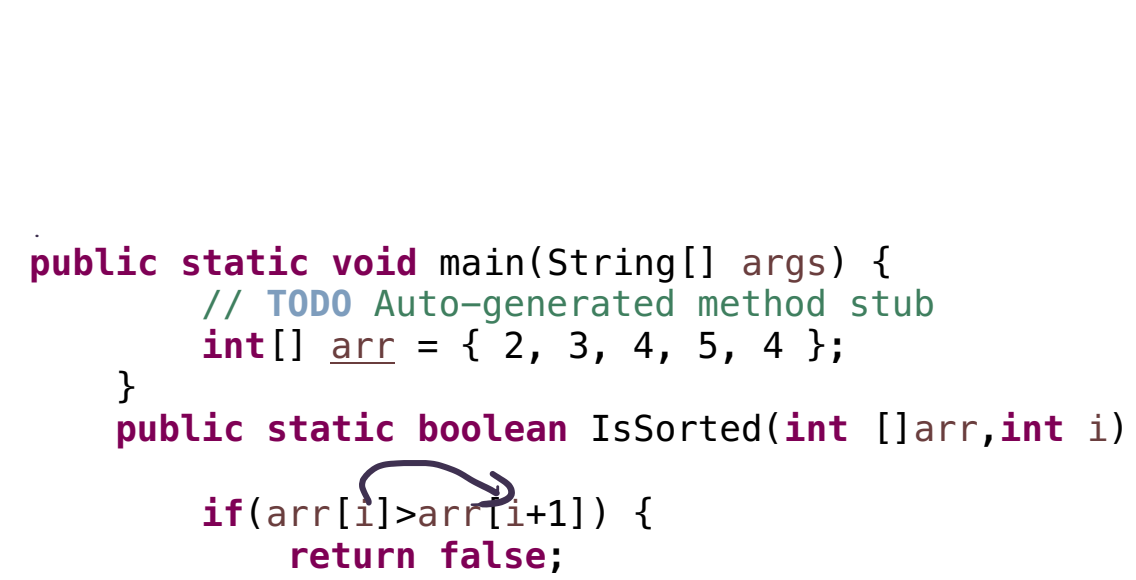
```java
int[] arr = { 2, 3, 4, 5, 4 };
```

0 1 2 3 4

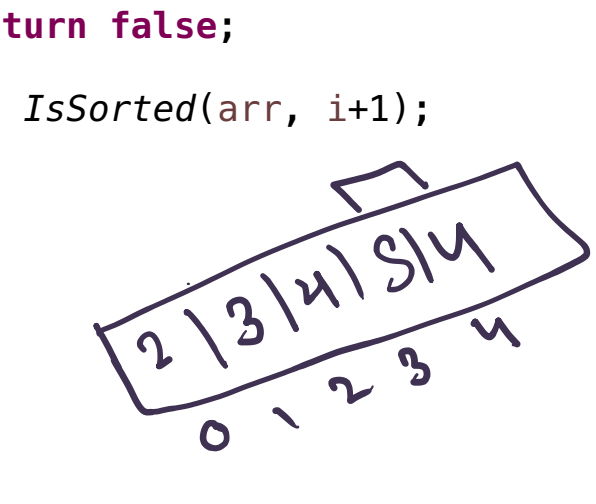
i=0 { if(arr[0] > arr[i+1]){
return false
}
arr, i+1

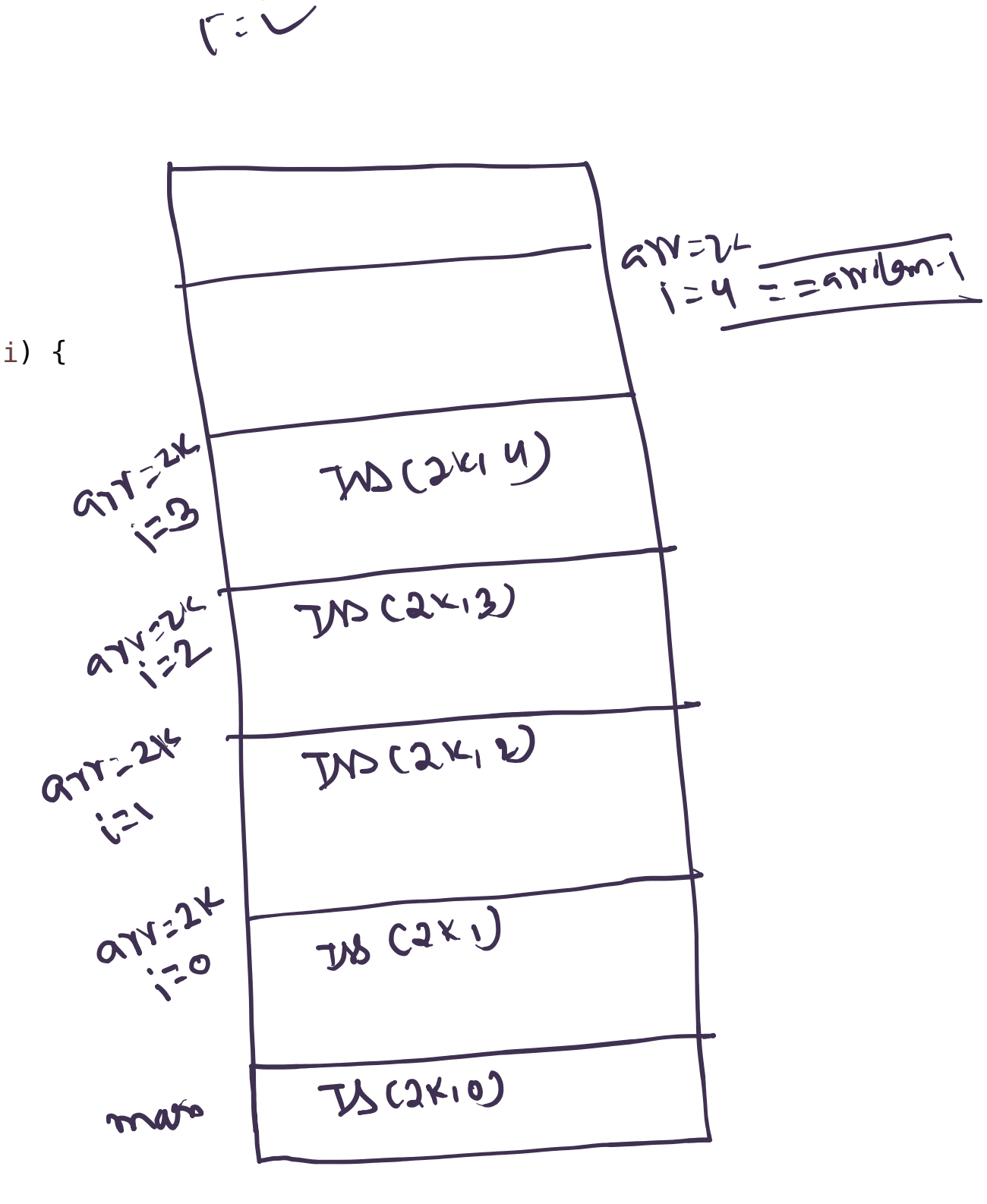
i=1 → if(arr[1] > arr[2]){
}

i=4

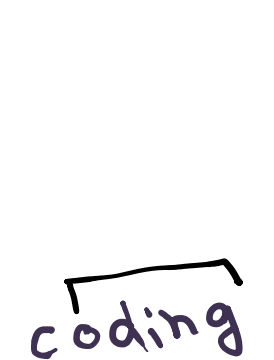


```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int[] arr = { 2, 3, 4, 5, 4 };
    }
    public static boolean IsSorted(int []arr,int i) {

        if(arr[i]>arr[i+1]) {
            return false;
        }
        return IsSorted(arr, i+1);
    }
```



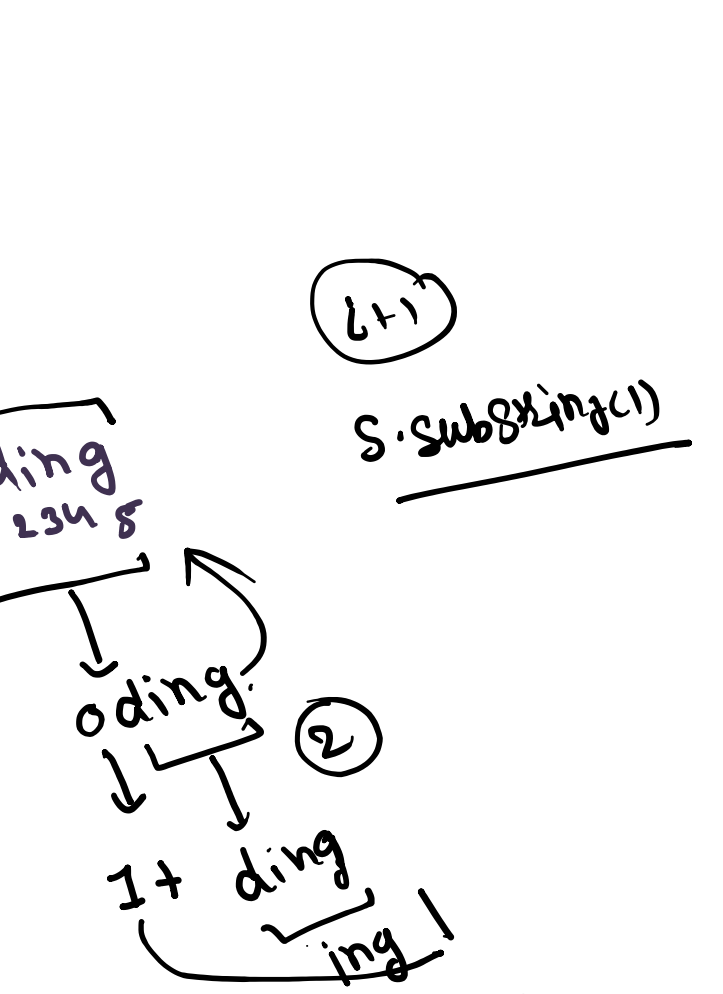


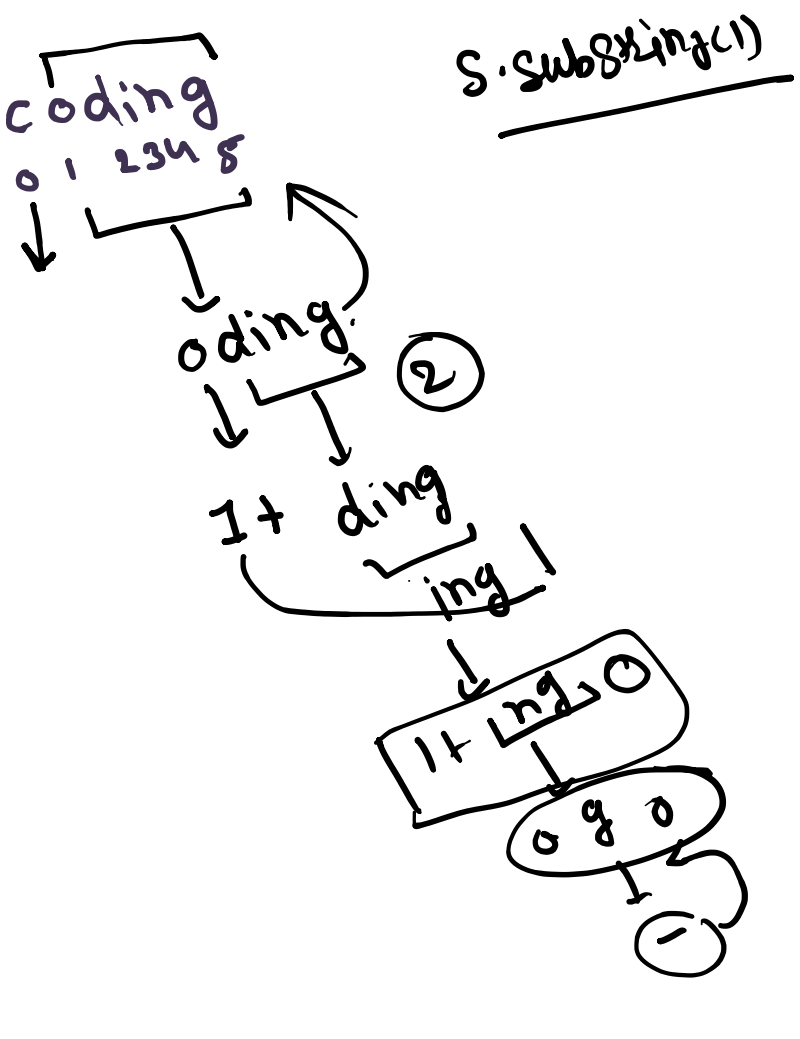
head tail



```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        System.out.println(fact(n, 1));
    }

    public static int fact(int n, int ans) {
        if (n == 0) {
            return 1;
        }

        return fact(n - 1, ans * n);

    }
```

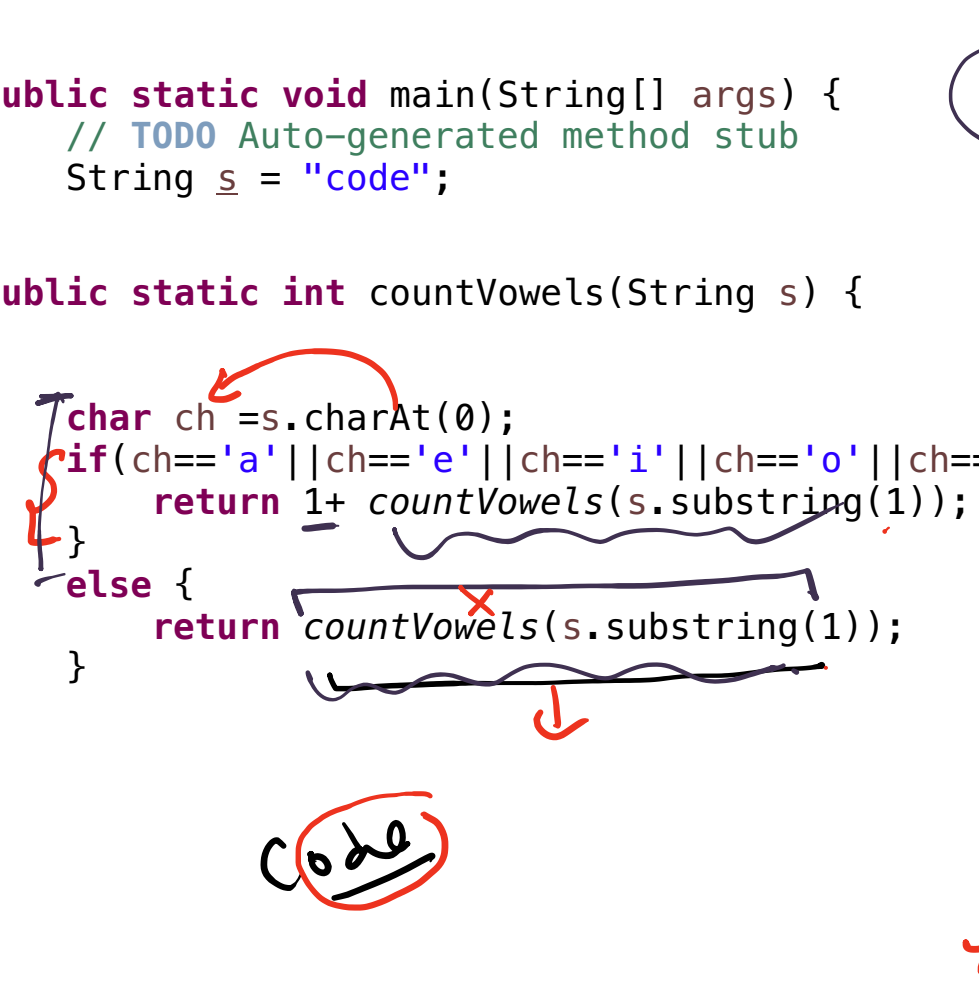
Pow(a, n-1, ans×a)





(i+1)

s.substring(1)



```java
public static void main(String[] args) {
        // TODO Auto-generated method stub
        String s = "code";
    }

    public static int countVowels(String s) {


        char ch =s.charAt(0);
        if(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u') {
            return 1+ countVowels(s.substring(1));
        }
        else {
            return countVowels(s.substring(1));
        }

    }
```

(code)